

PRISM

Platform for Research in Intelligent Synthesis of
Materials

AI-NATIVE AUTONOMOUS MATERIALS DISCOVERY

ESA SPARK PRIME CONTRACTOR • ITER SUPPLIER

The materials innovation cycle is fundamentally broken.

AI has identified over 2.2 million potential stable compositions, but only 736 have been verified in physical labs. Experimental validation cannot keep pace with computational prediction.

10-20

YEARS TO MARKET

2.2M

PREDICTED - ONLY 736 VERIFIED

€49B

AEROSPACE MATERIALS TAM

Replace trial-and-error with a deterministic, **self-optimizing loop.**

PRISM evaluates millions of compositions in minutes computationally, shifting failure from the physical lab to the digital domain. De-risking the discovery of Refractory High-Entropy Alloys for liquid rocket engine preburners.

TARGET

RHEAs for
preburners

REPLACES

Monel K500
legacy alloys

METHOD

Dual-loop:
compute + physical

Four Integrated Modules

Dual-loop workflow: computational inner loop for rapid optimization, physical outer loop for validation.



- **The Evolver (ACE Brain)**

Strategic reasoning engine implementing Agentic Context Engineering. Generator proposes, Reflector analyzes, Curator maintains the discovery playbook.

- **Mutator Fleet**

Specialized sub-agents using GFlowNet and flow matching to generate diverse candidates across the entire Pareto front.

- **Evaluator & A-Lab**

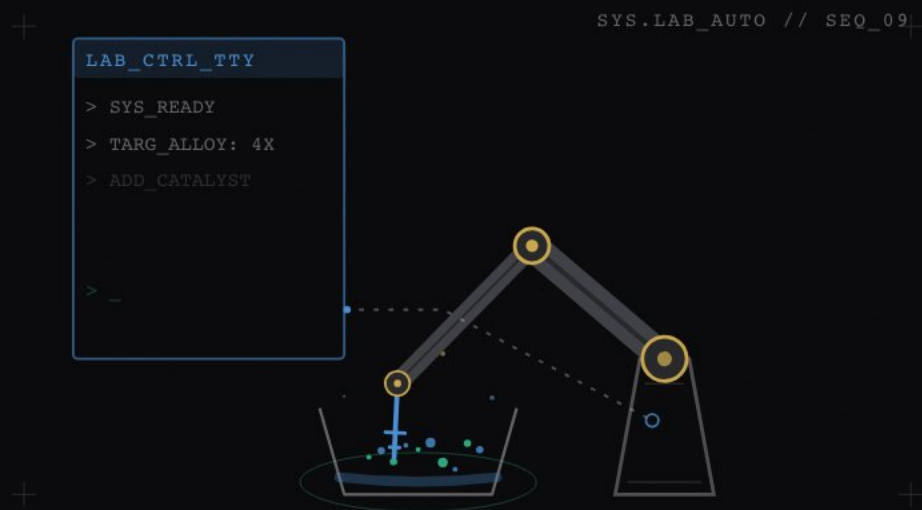
Multi-tiered fitness scoring: GNN surrogates (MACE-MH-1), DFT/CALPHAD checks, then robotic synthesis for ground-truth validation.

- **Materials Knowledge Graph**

Dynamic external brain ingesting literature, patents, and instrument data. Triggers information-seeking loops when search stagnates.

Autonomous Laboratory

Closing the loop: the real-world reward signal that trains the AI.



01 Recipe Translation

AI outputs target recipes: precursors, temperatures, thermal profiles.

02 Robotic Synthesis

Automated arms handle powder dosing and move samples through heating profiles.

03 Automated Characterization

System extracts raw diffraction patterns from synthesized material.

04 Phase Identification

Neural networks compare patterns against structure databases to confirm phases.

05 Feedback Loop

Empirical data structured via amsight and fed back into the Evolver's Reflector.

Not a whitepaper. Contracted and delivering.

VALIDATION

ESA SPARK Prime Contractor

FIRST!/FLPP initiative. Full AI discovery loop for first campaigns.

ITER Supplier

Titanium first-wall materials and rhodium targets for fusion.

ArianeGroup Interest

Direct end-user buy-in to test for full-flow staged combustion engines.

IP & CONSORTIUM

Proprietary IP Portfolio

GFlowNet sampling + CALPHAD thermodynamics as soft context. Enables discovery of metastable alloys. Patent filings in progress.

PARTNERS

Bimo Tech

AI Lead

Fraunhofer IAPT

AM Calibration

ArianeGroup

Requirements + Testing

amsight

Data Structuring



Total Addressable Market

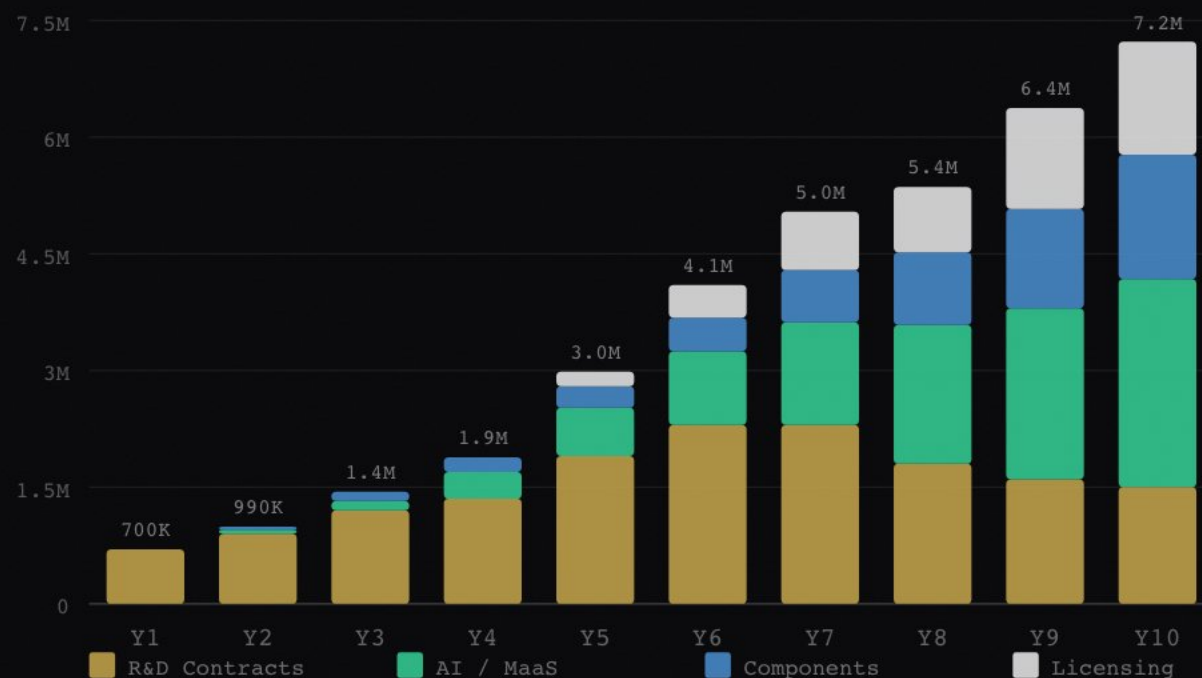
Aerospace extreme-environment materials

FOUR COMPOUNDING REVENUE STREAMS

R&D Contracts ESA, GSTP, Horizon Europe funding for technology maturation. Y1 €700K	RHEA Components Prototype coupons and qualified production parts for OEMs. Y2 €40K	AI Consulting / MaaS Discovery-as-a-Service for aerospace and energy OEMs. Y2 €50K	Platform Licensing Framework licenses and annual maintenance fees. Y5 €190K
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Bottom-Up Revenue Model

EBITDA-positive from Year 1. Improving unit economics as platform scales.



€1.8M EBITDA by Y10
25% margin at scale, positive from Year 1

€1.5M Net Income Y10
After 23% effective tax rate

35% Recurring by Y10
MaaS subscriptions + maintenance fees

38 mo Runway
€15.4M cumulative cash by Year 10

€5M

Seed Round

ALLOCATION

AI Team Scale-Up

ML engineers + materials scientists

40%

Autonomous A-Lab

Robotic synthesis & characterization hardware

30%

Compute Infrastructure

Dedicated GPU cluster for training

15%

IP & Market Dev

Patents, BD, conference presence

15%

ROADMAP TO AUTONOMY

I Computational Validation 6 mo

Full AI discovery loop running first SPARK campaigns.

II Hybrid Loop 12 mo

Integration with Fraunhofer IAPT additive manufacturing data.

III Full A-Lab Integration 24 mo

Direct instrument control adapters for completely closed-loop.



PRISM IS A SPIN-OFF
OF BIMO TECH

The Operating System for Accelerated Innovation.

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TECHNOLOGY CONCEPT BY

